Error Analysis

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Teach a useful hypothesis as an unanswered evidence question.

Supply case-level fixtures, explicit confounding and a complete follow-up protocol while separating supplied outcomes from executed model changes.

Keep one habit: Require all group/total directions and preserve development exposure before final evaluation.

MAIN LESSON · SECTION 1

Prepare the connected experiment

> Keep prior exposure, shared practice and independent evidence distinct.

Goal and main teaching plan

Primary target: Categorize development errors using **evidence**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Materials and prerequisite

Teach two 45-minute sessions. This week focuses on **error analysis** and controlled follow-up. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

It counts outcome combinations but does not explain the causes of individual errors.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Optional Python 3 editor; no external packages or AI accounts required

Opening retrieval

Without opening the old lesson, explain one key idea from Precision and Recall. Give an example and a mistake that the idea helps you avoid.

Look for: Choose the denominator that answers the question

Reference example, not the only acceptable wording: Teacher explanation: Predicted positives TP+FP=7; reference positives TP+FN=4.

Without opening the old lesson, explain one key idea from Confusion Matrix. Give an example and a mistake that the idea helps you avoid.

Look for: Build a labelled confusion matrix from matched reference and **prediction** pairs

Reference example, not the only acceptable wording: Reference clear/predicted clear contains V1,V2: 2. Clear/blocked contains V3: 1. Blocked/clear contains V4,V5: 2. Blocked/blocked contains V6,V7,V8: 3. Matrix [[2,1],[2,3]].

Without opening the old lesson, explain one key idea from Nearest Neighbors. Give an example and a mistake that the idea helps you avoid.

Look for: Compute absolute distances and rank **training** neighbours

Reference example, not the only acceptable wording: Distances to A,B,C,D,E are 1,1,5,7,9 cm. A and B are equally far from 3 on the numeric feature; their labels do not enter the distance calculation.

End goal

By the end, I can explain **error analysis** using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Precision and Recall

Confusion Matrix

Nearest Neighbors

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A model fails 6/10 backlit and 2/20 front-lit photos. Compute error rates and design a follow-up that tests lighting rather than leaf identity.

Concept to explain

Grouping errors by observable conditions connects a metric problem to possible data, label, feature, or model causes.

Term	Meaning
error slice	subset sharing a condition
failure cluster	related mistakes with a likely cause
root-cause hypothesis	testable explanation for errors
controlled repair	one change aimed at one cause

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach association, replay and controlled follow-up separately

> Largest error counts need their slice denominators and competing explanations.

Main supplied aggregate example and case-table context

Error analysis begins with **prediction** IDs and reference labels. Inspect features, collection conditions and the labelling policy. Do not assume every surprising **prediction** proves the reference is wrong.

Use development errors to propose a change and specify what outcome would support the explanation.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **error analysis** and controlled follow-up.

A validation audit has 12 dim-light cases with 4 errors and 18 bright-light cases with 3 errors. Rates are 4/12=1/3 and 3/18=1/6. Dim lighting is a useful lead, not proof of causation: camera, route or labels may also differ.

A planned change uses a **training**-fitted brightness normalization while keeping split and classifier settings fixed.

Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

Term	Meaning
Error analysis	Inspection of mistakes to propose explanations
Hypothesis	A testable proposed explanation
Controlled comparison	A comparison that changes one planned factor while holding relevant conditions fixed
Regression	Here: a new or worsened failure after an update. Next week the same word names numerical prediction, a different meaning

Error Clinic source is a teacher-authored fictional development replay: twelve D cases are dim/camera A/North and eighteen B cases bright/camera B/South. These conditions co-vary, so their slice difference cannot isolate lighting. D01–D04 are old errors; D01–D02 remain new errors. B01–B03 are old errors; B01–B04 are new errors.

Old and new predictions are supplied fixtures, not the result of a model execution in this lesson. A planned training-fitted brightness change is a hypothesis; its training data and implementation are not supplied, so no fitting or causal success is claimed here.

Trace the audit and causation question

Error analysis and controlled follow-up at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Inspect	IDs and conditions	Propose a hypothesis
Change	One planned factor	Keep comparison fixed
Evaluate	All validation records	Report gains and regressions

After the change, the same validation groups have 2/12 and 4/18 errors. Total errors fall from 7 to 6, but bright-light errors increase from 3 to 4. Report both results. A net improvement does not mean every part improved, and repeated experiments can overfit validation.

Does the difference prove dim light causes model errors? Explain the steps or evidence behind your response.

Read separate pictured14-case facts

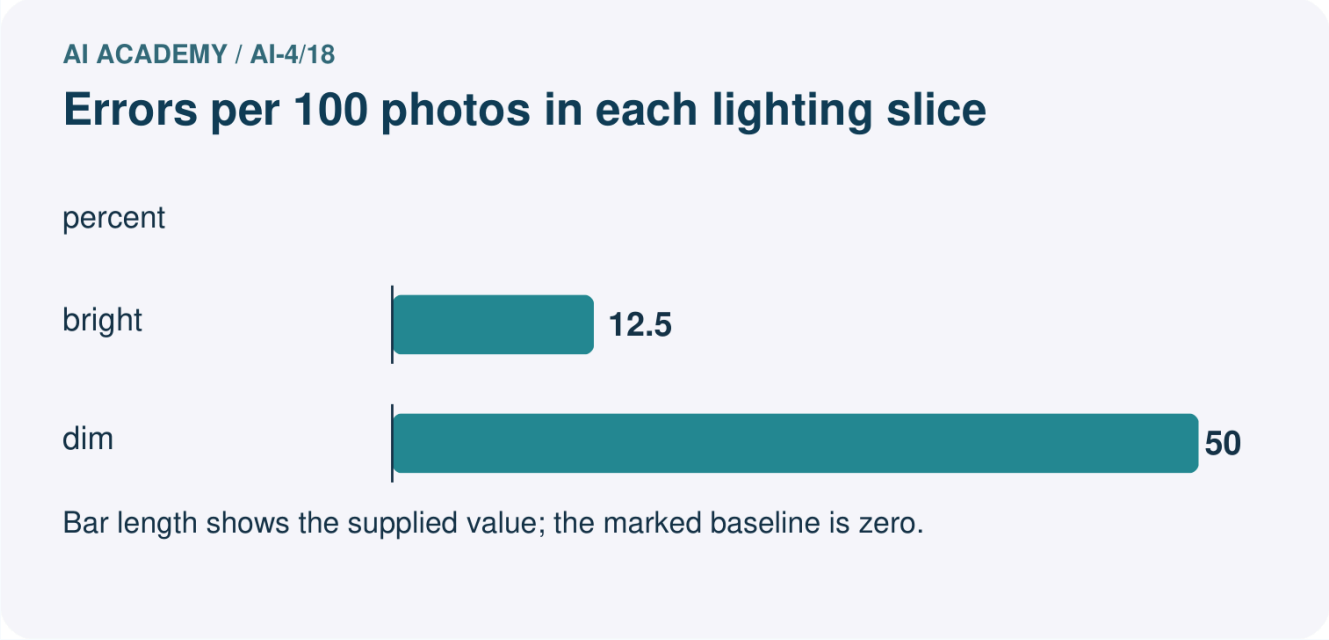
Find an error cluster before choosing a controlled follow-up.

Candidate errors: bright-light photos1/8; dim-light photos3/6. Images and reference labels are fixed for this audit.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

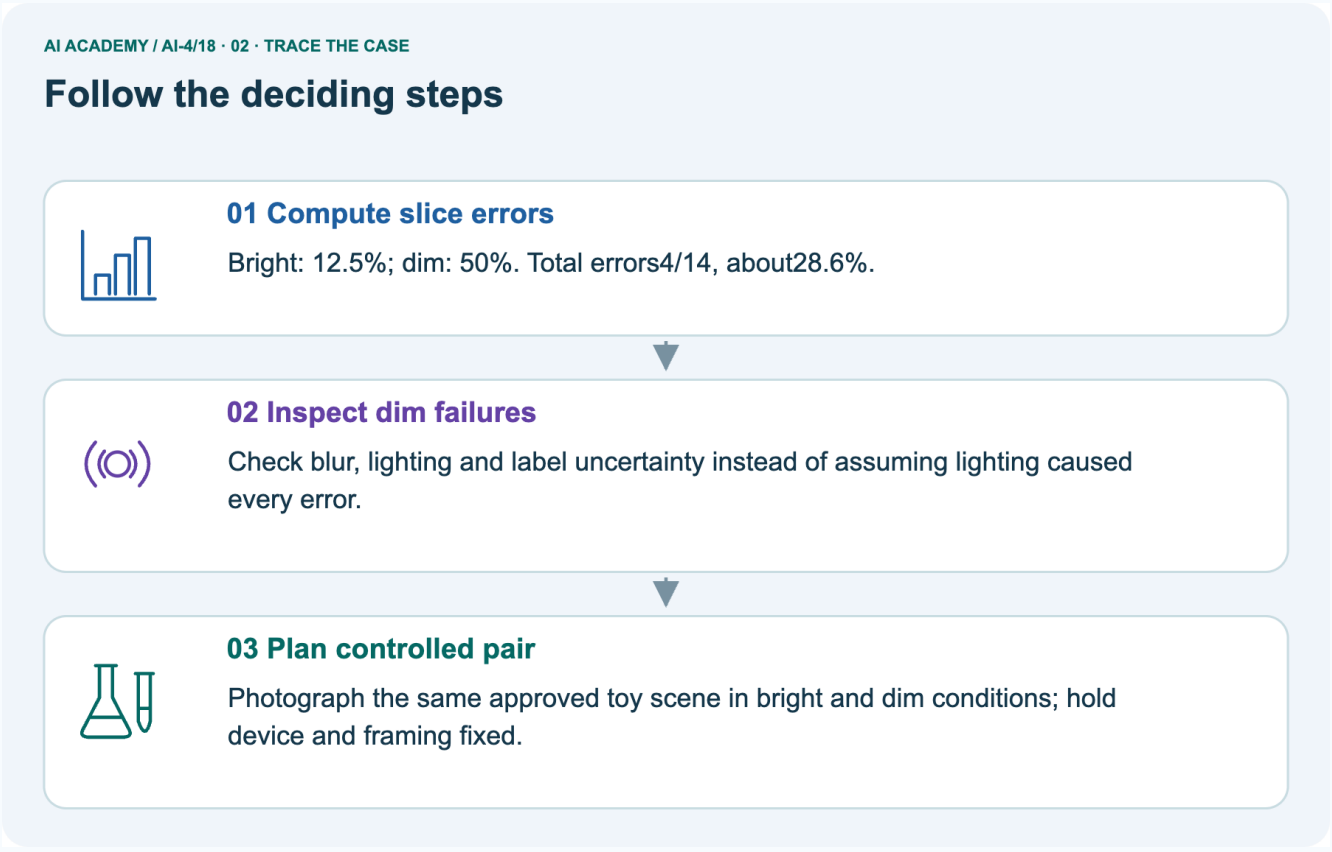
The main fixture has dim errors 4/12 and bright errors 3/18, later supplied as 2/12 and 4/18. Dim improves while bright worsens; total falls 7/30 to 6/30. The separate picture has bright 1/8 and dim 3/6, total 4/14. Its bar labels 12.5 and 50 are percentages, not counts from a hundred measured photos. The pictured same-case replay changes dim to 1/6; it is reused development evidence, not fresh generalization.

None of these numbers alone proves lighting caused the errors.



Follow the labelled picture with the matching explanation. Open the vector diagram

Trace percent bars and same-case replay limits



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute slice errors	Bright: 12.5%; dim: 50%. Total errors4/14, about28.6%.
2. Inspect dim failures	Check blur, lighting and label uncertainty instead of assuming lighting caused every error.
3. Plan controlled pair	Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

A slice difference suggests investigation; it does not establish a causal lighting explanation by itself.

Change: Dim errors fall to1/6 in a later same-case replay.

Prediction: The replay's dim error rate is 16.7%; it is development **evidence** if those cases guided the revision.

Reason: Observed improvement on reused cases is not an untouched final generalisation test.

▼ Additional backlit/ordinary comparison

Four errors occur among five backlit photos; two among twenty ordinary photos.

Compare rates and investigate.

Backlit error=**80%**; ordinary=10%. Inspect image quality and confounding differences. Association motivates controlled testing; it does not alone prove cause.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Cluster mistakes and propose cause-specific next tests.

MODEL IDEA Grouping errors by observable conditions connects a metric problem to possible data, label, feature, or model causes.

PROOF NEEDED Use an unseen case, visible calculation, **error analysis**, and a stated limitation.

UNDER THE HOOD Twelve errors are grouped: 6 low light, 3 blurred, 2 label disagreements, 1 unusual species. Largest cluster suggests a lighting representation problem, not proof of the cause.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

STEP 1 Twelve errors are grouped: 6 low light, 3 blurred, 2 label disagreements, 1 unusual species.

STEP 2 Largest cluster suggests a lighting representation problem, not proof of the cause.

STEP 3 Collect or transform low-light examples, changing one factor.

STEP 4 Rerun the same evaluation and inspect other slices for **regression**s.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

Exact explanation to model

Explain the added worked example

Key terms

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

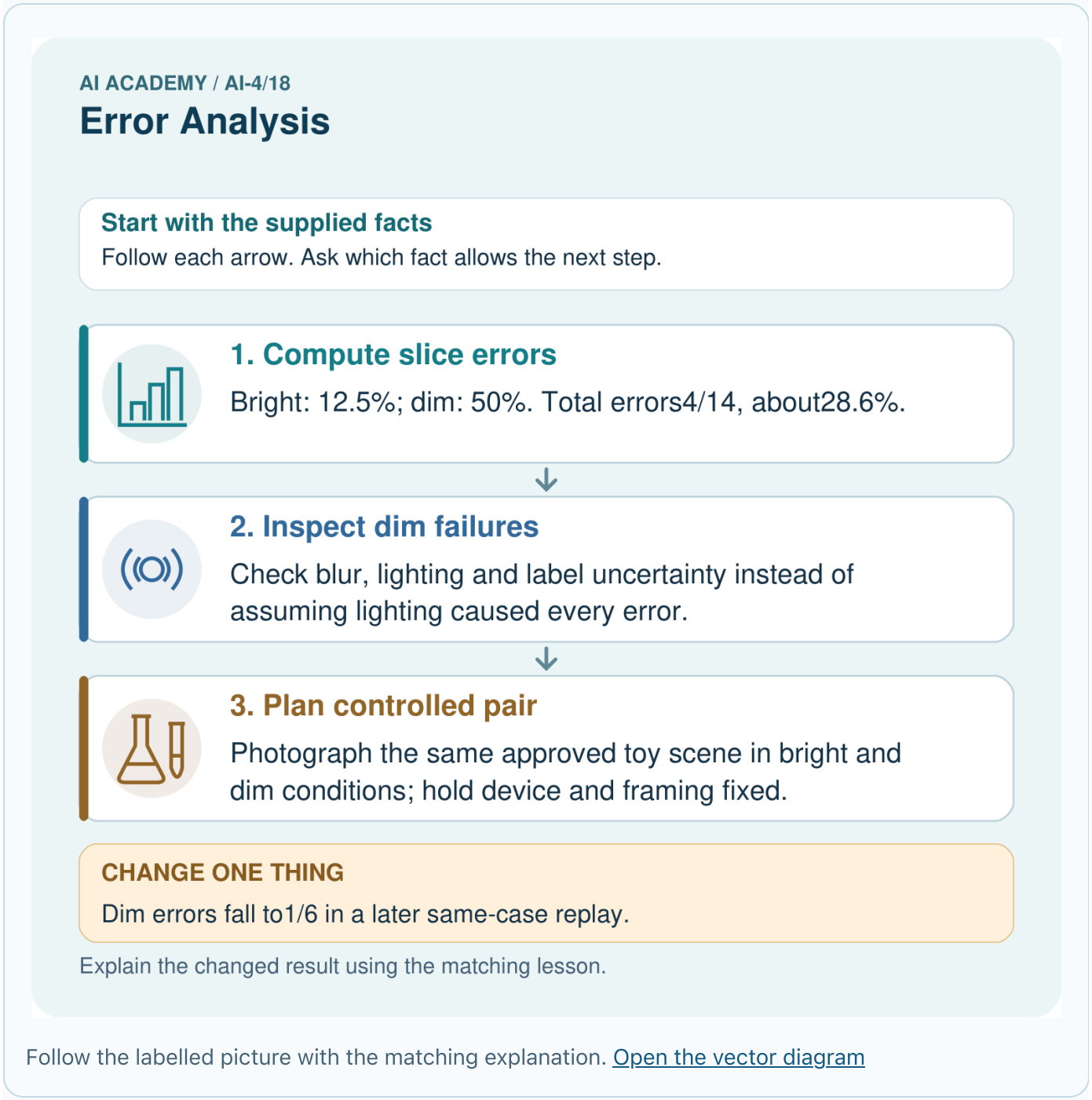
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Error Analysis

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Guide the calculation, then remove help

> Collect each learner's prediction before showing its key.

Current completion task and checked key

Completion task: Does the difference prove dim light causes model errors?

Teacher check: Teacher explanation: No. Other conditions and label quality may differ; it motivates a **hypothesis**.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Error analysis** and controlled follow-up. Proposed explanation: "A final answer is enough". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: Dim 4/12 about 33.3 percent; bright 3/18 about 16.7 percent.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case reasoning

What **evidence** would support the claim that a change helps fresh dim-light images? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Freeze the revision and test new appropriately labelled dim-light examples, retaining plant groups and comparable conditions.

Changed-case check: The replay's dim error rate is 16.7%; it is development **evidence** if those cases guided the revision. Observed improvement on reused cases is not an untouched final generalisation test.

Changed-case reasoning: The replay's dim error rate is 16.7%; it is development **evidence** if those cases guided the revision. Observed improvement on reused cases is not an untouched final generalisation test.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Demonstrate and guide practice

Board sequence

Write the given facts: Four errors occur among five backlit photos; two among twenty ordinary photos.

Write the question to explain: Compare rates and investigate.

Reveal after **prediction**s: Backlit error=**80%**; ordinary=10%. Inspect image quality and confounding differences. Association motivates controlled testing; it does not alone prove cause.

Use the earlier transfer question as additional practice. Reserve the selected fresh route A/B before-after error comparison for the first individual attempt and record support used.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided response: No. Other conditions and label quality may differ; it motivates a **hypothesis**. Exit response: The models are evaluated on different selected records.

Apply one documented review policy and compare both on the same versioned set.

Case-specific teaching sequence

Stage · Teacher action

> **Stage:** Session 1 start to 7 minutes
>
> **Teacher action:** Use the prerequisite question: What does a confusion matrix omit? Check the explanation before proceeding.

Stage: 7 to 18 minutes

Teacher action: Read the opening and model the worked evidence. Ask students to identify error analysis in context.

Stage: 18 to 32 minutes

Teacher action: Complete W 1 and W 2 with support: find the rates; explain the limit. Require the evidence, not only the final response.

Stage: 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Trace the Python check after the paper calculation.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between error analysis and hypothesis. Invite a counterexample to an overconfident claim.

Stage: 8 to 20 minutes

Teacher action: Discuss the application and complete W 3: Summarize the changed results without hiding a regression.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before giving hints. Assess these outcomes: Categorize development errors using evidence; Design a controlled change with a stated hypothesis. Record any support used.

Stage: 35 to 45 minutes

Teacher action: Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Keep predictions, observed or supplied outcomes and unmeasured cases separate.

Supplied inputs and checked guidance

Main total improves7→6 but bright worsens3/18→4/18; inspect B04 and competing device/route explanations. Counts supplied, no actual normalization model fit. Record fresh paired-scene protocol and withhold its nonexistent results. Independent route A5/10→3/10 improves50%→30%;B2/20→5/20 worsens10%→25%;total7/30→8/30 worsens.

Investigate B, retain the old comparison and do not claim overall improvement or a proved cause.

Predict, investigate and record

1. 1. Write the **hypothesis** and expected **evidence** before running.
2. 2. Error Clinic: annotate every wrong **prediction**, cluster by condition, write hypotheses, and select one measurable next test.
3. 3. Run one normal, one boundary, one missing, and one shifted case.
4. 4. Calculate the metric from raw counts or row-level errors.
5. 5. Interpret what changed, what did not, and what remains unsafe or unknown.

Keep the experiment ledger

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the current misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: Dim 4/12 about 33.3 percent; bright 3/18 about 16.7 percent.

A stronger claim makes a stronger project: The models are evaluated on different selected records. Apply one documented review policy and compare both on the same versioned set.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: Total worsens 7/30 to 8/30. A improves 50 to 30 percent, B worsens 10 to 25 percent.

Investigate B and keep the original as a comparison; do not claim overall improvement.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows	Next teaching action
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

Make the lab runnable

Use the supplied complete investigation for shared practice. Reserve the selected fresh route A/B before-after error comparison for the individual first attempt before hints or feedback; earlier transfer questions remain additional practice.

Diagnose and reteach

MISCONCEPTION TO REPAIR Looking closely at one surprising error is enough to diagnose the model.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected experiment transfer

> Keep this key until after a first attempt; record any help and prior exposure.

Selected independent task and checked key

Selected case: Old errors are 5/10 on route A and 2/20 on B. New errors are 3/10 and 5/20. State total and group changes and a justified next step.

Teacher explanation: Total worsens 7/30 to 8/30. A improves 50 to 30 percent, B worsens 10 to 25 percent. Investigate B and keep the original as a comparison; do not claim overall improvement.

Judge trace, scope and support

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Find the rates

Calculate the original error rates by lighting condition.

Teacher explanation: Dim 4/12 about 33.3 percent; bright 3/18 about 16.7 percent.

W2 Explain the limit

Does the difference prove dim light causes model errors?

Teacher explanation: No. Other conditions and label quality may differ; it motivates a **hypothesis**.

W3 Track regressions

Summarize the changed results without hiding a **regression**.

Teacher explanation: Total errors 7 to 6; dim 4 to 2 improves, bright 3 to 4 worsens.

W5 Repair a misleading audit

A team removes all disputed records only from the new model report. Why is the comparison invalid?

Teacher explanation: The models are evaluated on different selected records. Apply one documented review policy and compare both on the same versioned set.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Backlit error=**80%**; ordinary=10%. Inspect image quality and confounding differences. Association motivates controlled testing; it does not alone prove cause.

Additional transfer practice question

Of eight errors, six are on blurry images and two on clear images. Can you conclude blurry images have a higher error rate without group totals?

Reasoned practice answer

No. The counts identify a useful investigation, but rates require the total number of blurry and clear examples. Inspect the images and compute each group's denominator.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “More errors in a group automatically means a higher error rate.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Rates are **60%** and 10%. Photograph comparable or paired leaves under controlled lighting while holding device and labels fixed, keeping identity groups out of **training**/test overlap.

The original association motivates the test but does not prove lighting caused every error.

Explanations for the additional workbook examples

W1 Find the rates

W2 Explain the limit

W3 Track regressions

W4 Apply independently

Old errors are 5/10 on route A and 2/20 on B. New errors are 3/10 and 5/20. State total and group changes and a justified next step.

Teacher explanation: Total worsens 7/30 to 8/30. A improves 50 to 30 percent, B worsens 10 to 25 percent. Investigate B and keep the original as a comparison; do not claim overall improvement.

W5 Repair a misleading audit

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit the experiment

> Retain failures and assess a fresh parallel case after help.

Project evidence

Create the Plant Classifier error log linking case ID, actual, predicted, confidence, condition, likely cause, and next action.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support and selected reassessment

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **error analysis** and **hypothesis** in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: Total worsens 7/30 to 8/30. A improves 50 to 30 percent, B worsens 10 to 25 percent. Investigate B and keep the original as a comparison; do not claim overall improvement.

Fresh retry: old groups 2/8 and 4/12; new 3/8 and 2/12. Total improves 6/20 to 5/20 while the first group regresses. Require both statements. Mark each criterion as independent, supported or not yet demonstrated.

Reassess the reasoning after feedback; do not replace the first attempt silently.

Delayed return

Suggested return points in this level: ai-4/19, ai-4/21, ai-4/24. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge and reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 19: **Regression** and Residuals. Retrieve today's **evidence** and identify what carries forward and what is new.

Next week: **Regression** and residuals. Carry forward the saved **evidence** and the distinction between development choices and justified claims.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Responsible Plant Classifier

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route without repeating the whole cycle.

Planning and assessment reference

Teaching target: Categorize development errors using **evidence**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Sections 1–6; record the brief recall in the workbook.

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>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Does the difference prove dim light causes model errors?

Reasoned practice answer: Teacher explanation: No. Other conditions and label quality may differ; it motivates a **hypothesis**.

Check every learner before discussion: Use the worked case for **Error analysis** and controlled follow-up. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Ask for the relevant counts, units, procedure or saved record. For this week: Dim 4/12 about 33.3 percent; bright 3/18 about 16.7 percent.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Dim errors fall to1/6 in a later same-case replay. Require a **prediction** and a reason before comparing.

Expected result: The replay’s dim error rate is 16.7%; it is development **evidence** if those cases guided the revision. Reason: Observed improvement on reused cases is not an untouched final generalisation test.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 8

Reference · Feedback and evidence records

> Keep supported rehearsal separate from independent performance.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ **Earlier wording, alternatives and picture transcripts**

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

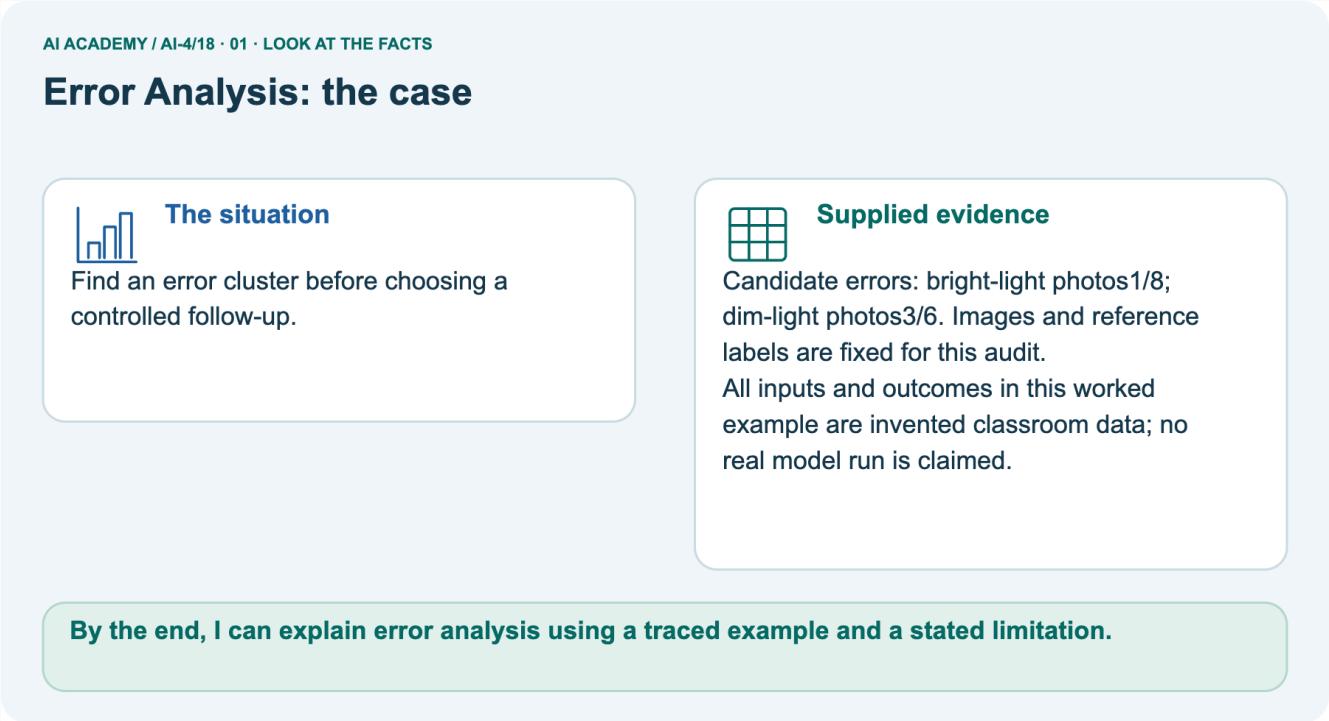
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a specific reasoning move.

01. Error Analysis: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

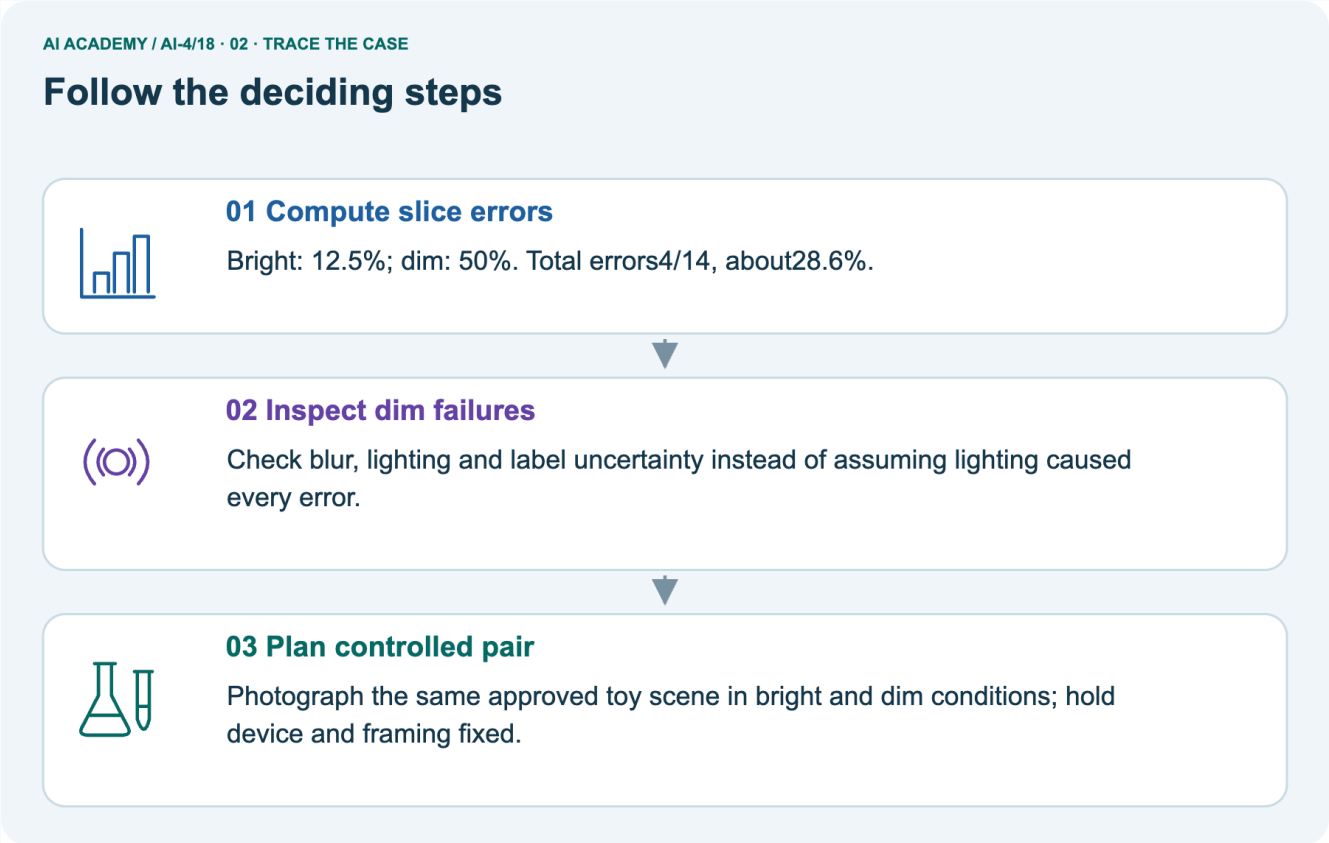
Find an error cluster before choosing a controlled follow-up. Supplied facts: Candidate errors: bright-light photos1/8; dim-light photos3/6. Images and reference labels are fixed for this audit.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

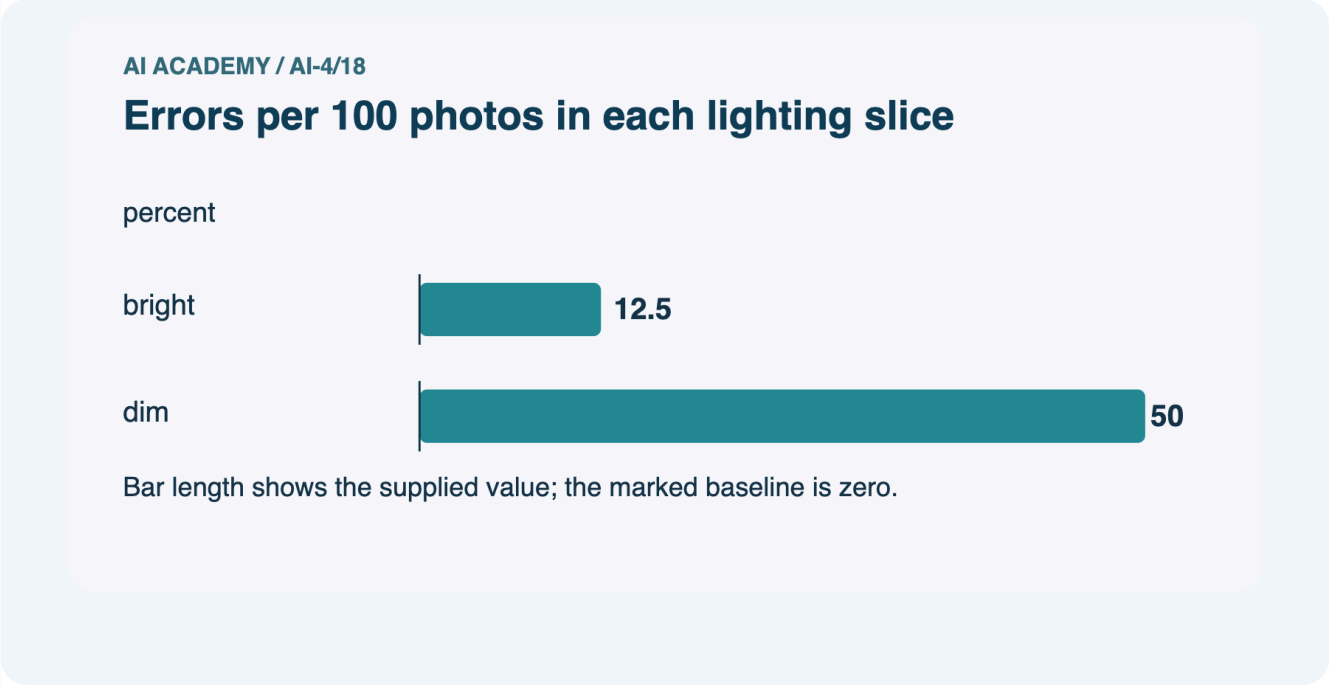
Compute slice errors: Bright: 12.5%; dim: 50%.

Total errors4/14, about28.6%. → Inspect dim failures: Check blur, lighting and label uncertainty instead of assuming lighting caused every error. → Plan controlled pair: Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Errors per 100 photos in each lighting slice



Follow the labelled picture with the matching explanation. Open the vector diagram

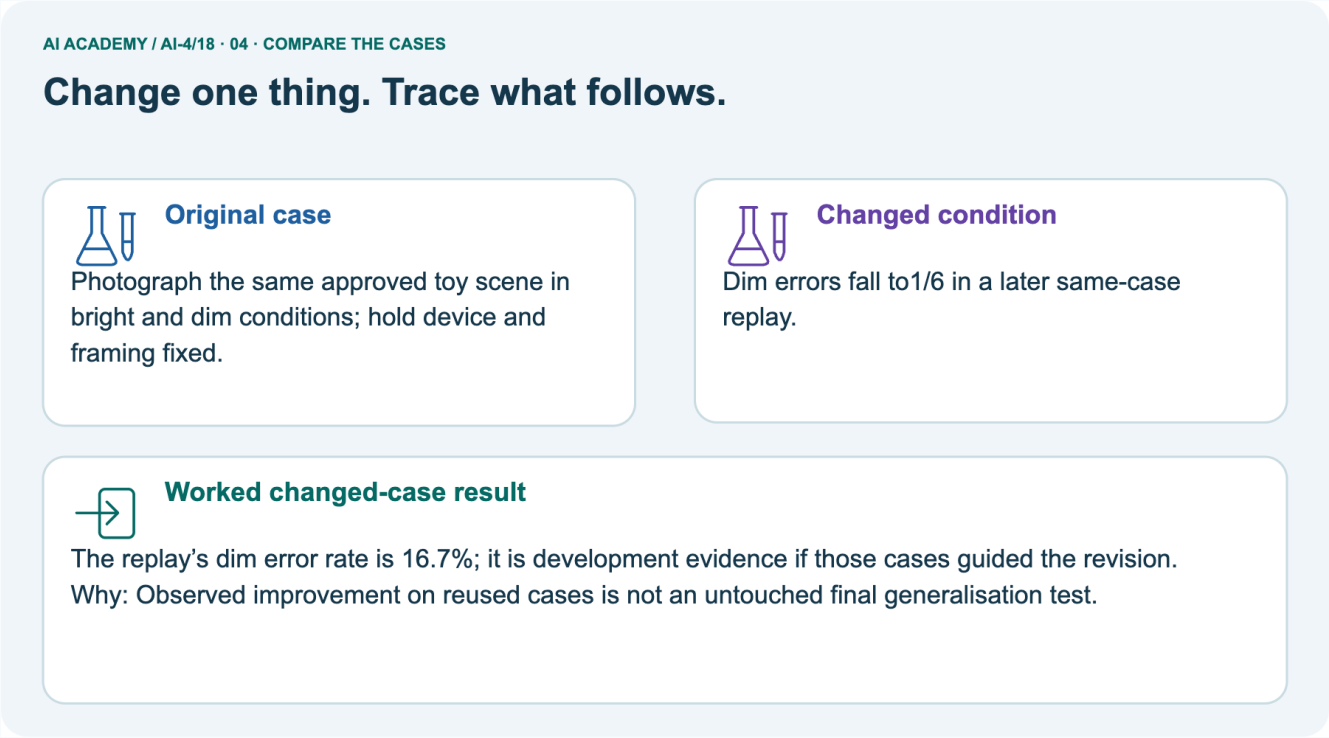
Errors per 100 photos in each lighting slice. Units: percent. bright: 12.5. dim: 50.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed. Changed condition: Dim errors fall to1/6 in a later same-case replay.

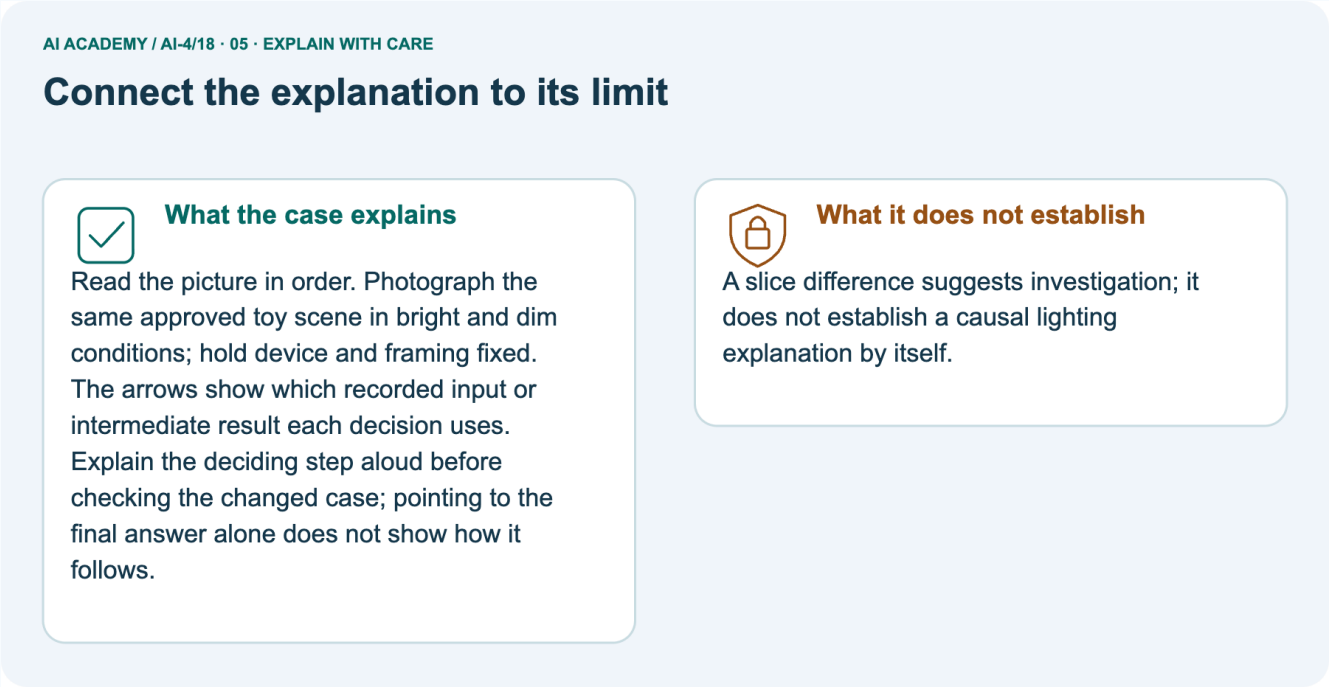
Expected result: The replay's dim error rate is 16.7%; it is development **evidence** if those cases guided the revision. Reason: Observed improvement on reused cases is not an untouched final generalisation test.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Observed improvement on reused cases is not an untouched final generalisation test.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

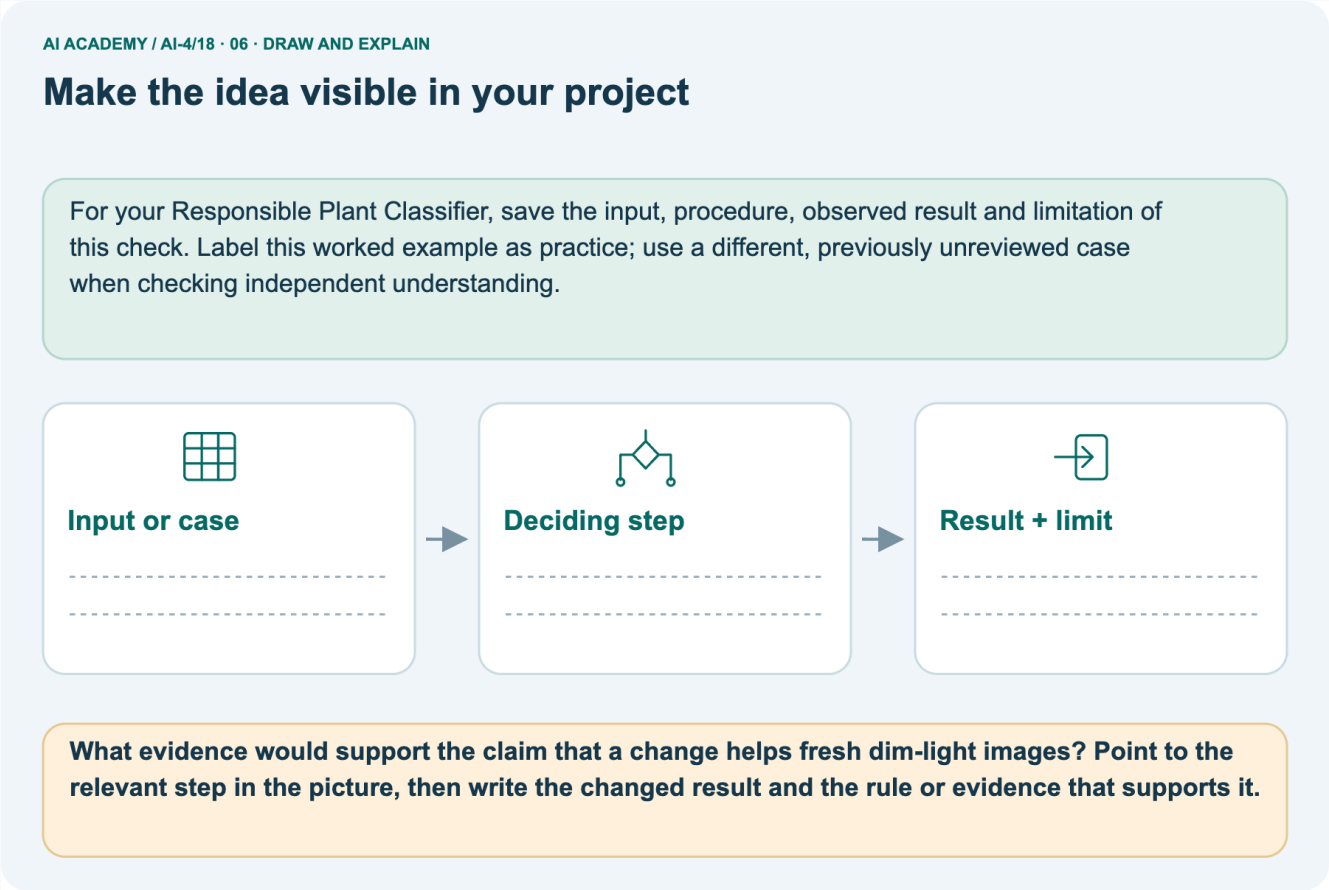
Limit: A slice difference suggests investigation; it does not establish a causal lighting explanation by itself.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A slice difference suggests investigation; it does not establish a causal lighting explanation by itself.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What **evidence** would support the claim that a change helps fresh dim-light images? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Freeze the revision and test new appropriately labelled dim-light examples, retaining plant groups and comparable conditions.

Changed-case check: The replay's dim error rate is 16.7%; it is development **evidence** if those cases guided the revision. Observed improvement on reused cases is not an untouched final generalisation test.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Error Analysis: the case

02. Follow the deciding steps

03. Errors per 100 photos in each lighting slice

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.